

MOMBETSU, Japan

WMO: 474350

Lat: 44.3452N

Lon: 143.3560E

Elev: 58

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.8	10.9	-0.1	5.5	9.9	2.7	6.3	13.0	24.4	26.1	21.6	25.0	7.2	210	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.0	82.0	69.5	78.2	68.1	75.0	67.2	72.6	79.2	70.7	76.0	68.9	73.5	8.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.3	111.6	76.3	68.6	105.3	74.5	66.9	99.2	72.6	36.4	79.6	34.7	76.1	33.2	73.6	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.5	17.8	15.5	DB	3.2	90.2	3.6	3.8	0.6	93.0	-1.5	95.2	-3.6	97.4	-6.2	100.2
			WB	2.4	75.6	3.5	2.0	-0.2	77.0	-2.2	78.2	-4.2	79.3	-6.7	80.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	21.9	21.8	29.5	40.4	49.6	55.6	63.2	67.4	62.3	50.9	38.3	27.0
	DBStd	17.06	5.47	6.48	6.59	7.43	8.47	6.88	6.54	5.68	5.33	5.77	7.16	5.64
	HDD50	3863	872	790	635	304	114	21	0	0	0	59	355	713
	HDD65	7842	1337	1210	1100	737	481	291	117	38	115	438	800	1178
	CDD50	1715	0	0	0	17	102	190	408	539	368	86	5	0
	CDD65	218	0	0	0	0	4	9	60	112	33	0	0	0
	CDH74	941	0	0	0	5	48	57	245	464	120	2	0	0
	CDH80	211	0	0	0	1	11	9	63	111	16	0	0	0
Wind	WSAvg	6.9	8.1	7.5	7.8	7.5	6.5	5.3	4.7	5.2	6.7	7.6	7.8	8.0
Precipitation	PrecAvg	33.7	1.7	1.3	1.4	1.9	2.4	2.6	4.3	4.7	5.4	3.0	2.6	2.5
	PrecMax	44.6	4.8	2.8	2.8	5.9	5.6	5.2	9.8	12.5	12.3	6.3	6.1	5.8
	PrecMin	21.3	0.8	0.5	0.4	0.3	0.7	0.9	0.7	0.7	2.1	0.8	0.7	1.1
	PrecStd	6.1	0.9	0.6	0.6	1.1	1.2	1.2	2.1	3.0	3.1	1.3	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	44.1	53.4	71.6	79.6	80.5	87.3	88.7	82.2	70.8	62.1	46.9
		MCWB	34.7	37.7	43.0	52.6	58.7	61.6	72.5	73.7	68.2	58.8	50.0	41.3
	2%	DB	34.2	38.4	46.4	62.7	72.7	74.0	80.3	83.0	77.4	66.5	56.7	41.5
		MCWB	30.7	32.7	38.5	48.6	56.0	62.1	70.0	71.8	66.1	56.3	48.0	36.9
	5%	DB	32.0	34.3	42.2	57.1	67.4	69.5	75.8	78.8	73.7	63.2	52.8	37.9
		MCWB	29.6	30.1	35.8	46.0	54.8	60.6	68.9	70.5	64.8	55.2	45.9	34.2
	10%	DB	30.2	31.1	38.9	52.3	62.6	65.8	72.5	75.6	70.8	60.5	49.2	35.5
		MCWB	27.7	28.0	33.6	43.3	52.7	59.0	67.3	69.2	63.5	53.4	43.9	32.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.2	37.6	43.6	54.1	61.0	66.3	74.7	75.4	71.6	61.5	52.6	42.1
		MCDB	39.7	44.0	52.7	66.4	74.2	75.7	81.8	84.9	78.9	66.7	59.6	46.0
	2%	WB	31.9	33.1	39.1	49.9	58.1	63.4	72.0	73.2	68.8	58.7	49.0	37.4
		MCDB	33.6	37.7	45.2	61.7	69.2	71.1	78.3	80.3	74.5	63.8	54.9	40.4
	5%	WB	29.8	30.4	36.5	46.6	55.7	61.5	69.6	71.6	66.5	56.5	46.5	34.7
		MCDB	31.8	33.5	41.6	55.7	65.6	68.0	74.6	77.2	71.3	61.5	51.9	37.5
	10%	WB	27.7	28.4	34.0	43.9	53.3	59.6	67.5	70.0	64.6	54.3	44.2	32.4
		MCDB	30.0	30.9	38.4	51.8	61.2	65.0	72.0	74.6	69.3	59.5	49.0	35.1
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.9	11.7	13.4	13.6	11.0	9.7	10.0	12.9	14.2	11.6	10.2
		MCDBR	12.7	16.9	17.9	22.9	23.4	18.0	15.9	15.5	16.6	17.8	18.3	13.7
		MCWBR	10.8	13.0	12.3	12.8	11.3	9.1	8.1	7.6	9.3	11.3	12.5	11.1
	5% WB	MCDBR	11.8	15.6	17.4	21.1	20.6	15.8	14.5	13.3	13.6	15.6	17.1	13.1
		MCWBR	10.7	12.4	12.5	12.8	11.0	8.8	8.2	7.1	8.6	10.9	12.3	11.2
Clear-Sky Solar Irradiance	taub	0.330	0.373	0.437	0.496	0.513	0.495	0.477	0.454	0.403	0.396	0.372	0.346	
	taud	2.071	1.892	1.827	1.853	1.916	2.065	2.184	2.266	2.358	2.227	2.185	2.093	
	Ebn at Noon	237	246	246	244	244	250	253	254	259	240	222	216	
	Edh at Noon	33	47	58	61	59	51	45	40	34	34	30	29	
All-Sky Solar Radiation	RadAvg	450	750	1110	1415	1530	1538	1449	1335	1133	814	479	361	
	RadStd	39	81	88	111	126	150	175	86	95	40	42	30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page